BOT CHAIN TESTNET · ERC-4337

Conform

A conformance workbench for account abstraction stacks. Probe the network, bundler, contracts and account — then prove a UserOperation end to end.

> “A submitted operation is not a confirmed operation.”

github.com/Fr3dy03/Conform On-chain records Download PDF

THE PROBLEM

Account abstraction fails in pieces.

- EntryPoint, factory and implementation drift silently across deployments.
- Bundlers omit methods, miscalculate verification gas, or reject estimates they just returned.
- Counterfactual accounts break gas estimation until someone remembers to prefund them.
- Receipts end in a hash — with no explanation of which layer actually failed.

Today’s workflow is paste-and-pray: explorer tabs, bundler logs, ad-hoc scripts — redone from scratch for every chain and every stack change.

THE SOLUTION

One page. Five stages. Twenty-two checks.

Network | Bundler | Contracts | Account | UserOperation

- Live probes: chain ID, client version, latency; bundler method surface and EntryPoint support.
- Bytecode-verified contracts: EntryPoint v0.7, SimpleAccountFactory, implementation lookup.
- SimpleAccount derivation from the owner EOA, funding state, EntryPoint nonce.
- Full path: estimate → sign the raw `userOpHash` → submit → verify the receipt.
- Sponsorship is reported honestly: unknown until a verified paymaster proves otherwise.

HOW IT WORKS

Keys stay in the wallet. Evidence comes from the chain.

BROWSER

Next.js console. Never receives keys. Injected wallet signs the raw hash via `personal_sign`; signature stays in memory.

SAME-ORIGIN RELAYS

`/api/rpc` & `/api/bundler`: method allowlists, 64 KB body cap, 180 req/min, 20 s / 30 s upstream timeouts.

BOT CHAIN TESTNET

`rpc.bohr.life` · `bundler.bohr.life` · EntryPoint v0.7 · SimpleAccountFactory.

Receipt rule: success = bundler receipt success: true AND outer transaction status 0x1.

Pure TypeScript core in `packages/chain` (viem + Zod) — derivation, diagnostics, hashing and reporting, testable without a browser (19 unit tests).

EVIDENCE MODEL

Three rules the UI cannot break.

1 — Submitted ≠ confirmed. A bundler hash is a lead, not a result. Success requires inner success=true and outer status 0x1.

2 — Sponsorship is unknown until proven. A configured paymaster proves nothing; only a real sponsored UserOperation does.

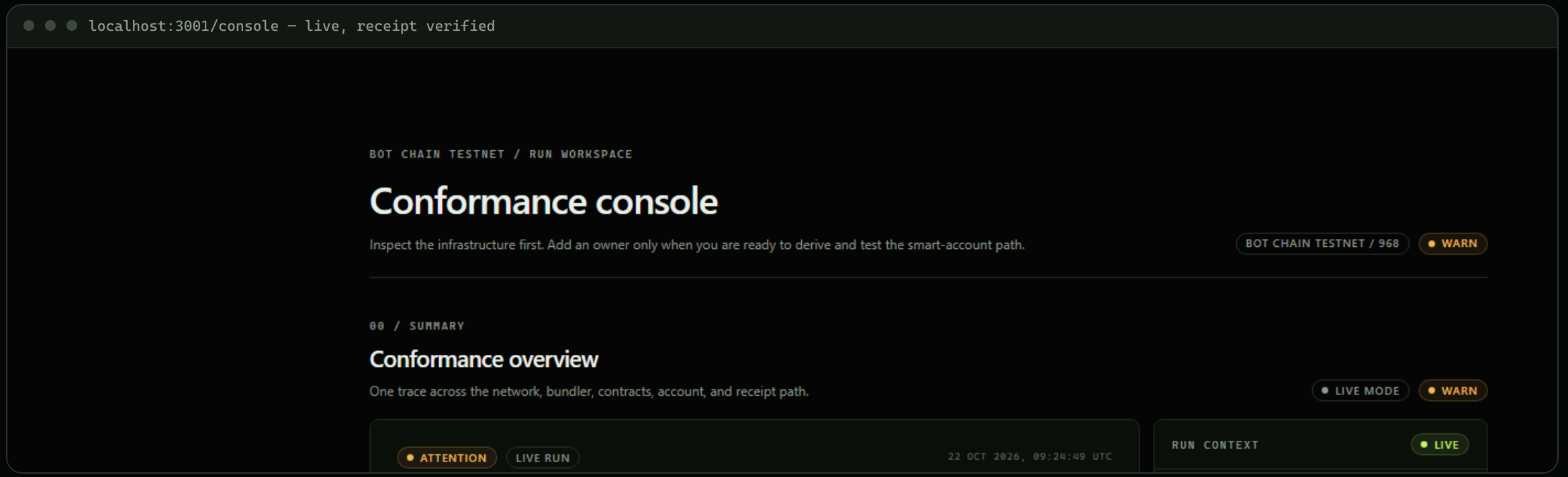
3 — Reports never carry secrets. No keys, seed phrases, signatures or raw payloads in reports or share links. /report is noindex; demo mode never connects, signs or broadcasts.

PROOF · LIVE TESTNET RUN

Headless end-to-end, receipts verified.

21	0	1	2
PASS	FAIL	UNKNOWN · SPONSORSHIP	USEROPS ON CHAIN

- Full UI flow driven in Chrome: connect → derive → sign → submit → receipt verified.
- Relay guards exercised: cross-origin 403, bad method 403, oversized 413, GET 405.
- Zero console errors; report export & share link contain no secrets.
- 22 Oct 2026 — run reproduced by the checked-in E2E harness.



ON-CHAIN INTERACTION RECORDS

Wallet explorer links — verify everything yourself.

INTERACTION	TRANSACTION	BLOCK	GAS USED	RESULT
SimpleAccount deployment	0xce8c0c20…b2995	27,678,217	275,279	0x1 · success
Conformance UserOperation	0x3769fc03…58d7d	27,735,482	87,757	0x1 · success

Owner EOA 0x5a043e…74FD37 Smart account 0x750c09…15A731 Full records

BOT Chain Testnet (chain 968). Both operations ran without a paymaster; gas prefunded from the smart account. Receipts re-verified over RPC.

SECURITY & STACK

Hardened by default, boring on purpose.

SECURITY POSTURE

- No key custody: the wallet signs; the signature never leaves browser memory.
- Relays allowlist methods — read-only RPC, bundler methods for the UserOp path only.
- Sanitized error text; reports exclude keys, signatures and raw payloads.
- Demo mode never connects, signs or broadcasts.

STACK

- Next.js 16 · React 19 · TypeScript 5.9 · Tailwind 4.
- viem · Zod · Vitest — pure-TS core, typed end to end.
- Monorepo: `apps/web` UI + relays, `packages/chain` logic, `packages/ui` components.
- Vercel-ready: root directory `apps/web`, zero env vars required.

LIMITATIONS & ROADMAP

Honest about what it proves.

TODAY

- BOT Chain testnet only; pinned to chain 968.
- Bundler checks are a sampled method matrix, not full compliance.
- Sponsorship unproven without a real paymaster operation.
- EOA paymaster and EIP-7702 flows out of scope.

NEXT

- Verified paymaster flows with real sponsored UserOperations.
- EIP-7702 delegation path on the same receipt discipline.
- Multi-bundler matrix and scheduled E2E runs in CI.
- Mainnet profile once BOT Chain mainnet endpoints stabilize.

RUN IT

Verify everything. Trust receipts.

```
npm install && npm run dev → localhost:3000/console · add ?demo=1 for labelled sample evidence
```

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Conform is a BOT Chain testnet workbench for ERC-4337 conformance. Testnet only. No keys, no custody, no claims without receipts.